

2021 Ph H2 Q12

Section: Electricity

Topic: Monitoring and Measuring a.c.

This question focuses on measuring alternating current using an oscilloscope.

(a)

State what is meant by the root mean square (rms) value of an alternating current

✔ The **rms current** is the value of **direct current** that would deliver the **same power** to a resistor as the alternating current does **on average**.

(b)

Calculate the rms voltage

Given:

- Peak voltage $V_{\text{peak}} = 8.5 \text{ V}$

Use the relationship:

$$V_{\text{rms}} = \frac{V_{\text{peak}}}{\sqrt{2}} = \frac{8.5}{\sqrt{2}} = 6.0 \text{ V}$$

(c)

Calculate the frequency of the alternating supply

From the oscilloscope:

- Timebase setting = 5.0 ms/div
- One full wave covers 4 divisions

$$T = 4 \times 5.0 \text{ ms} = 20.0 \text{ ms} = 20.0 \times 10^{-3} \text{ s}$$

$$f = \frac{1}{T} = \frac{1}{20.0 \times 10^{-3}} = 50 \text{ Hz}$$

Revision Tips

- $V_{\text{rms}} = \frac{V_{\text{peak}}}{\sqrt{2}}$ — use this for sinusoidal waveforms
- Timebase $\times$ divisions gives the **period** of the wave
- Frequency is the **reciprocal of period**: $f = \frac{1}{T}$
- RMS values represent **equivalent DC power**